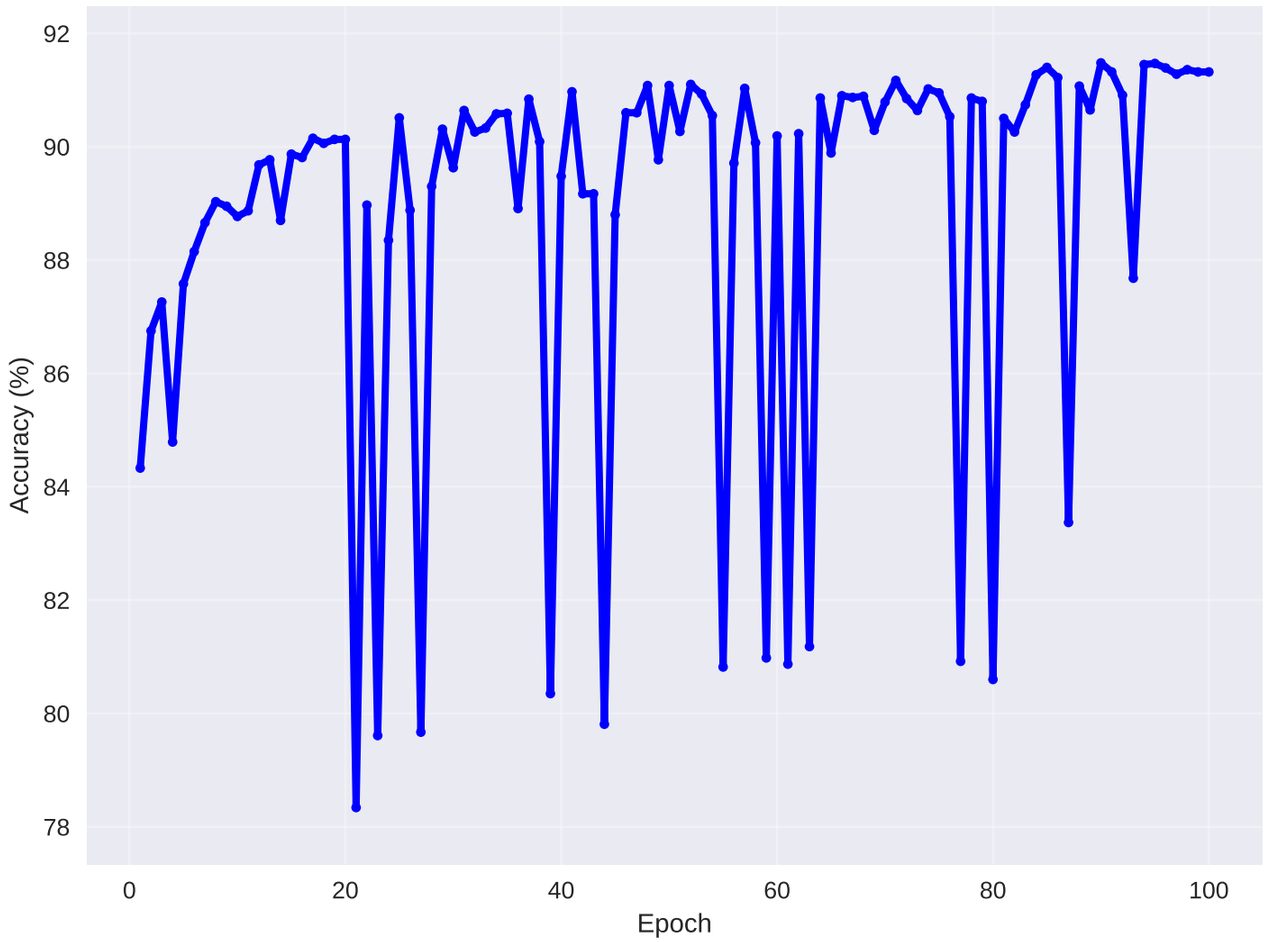
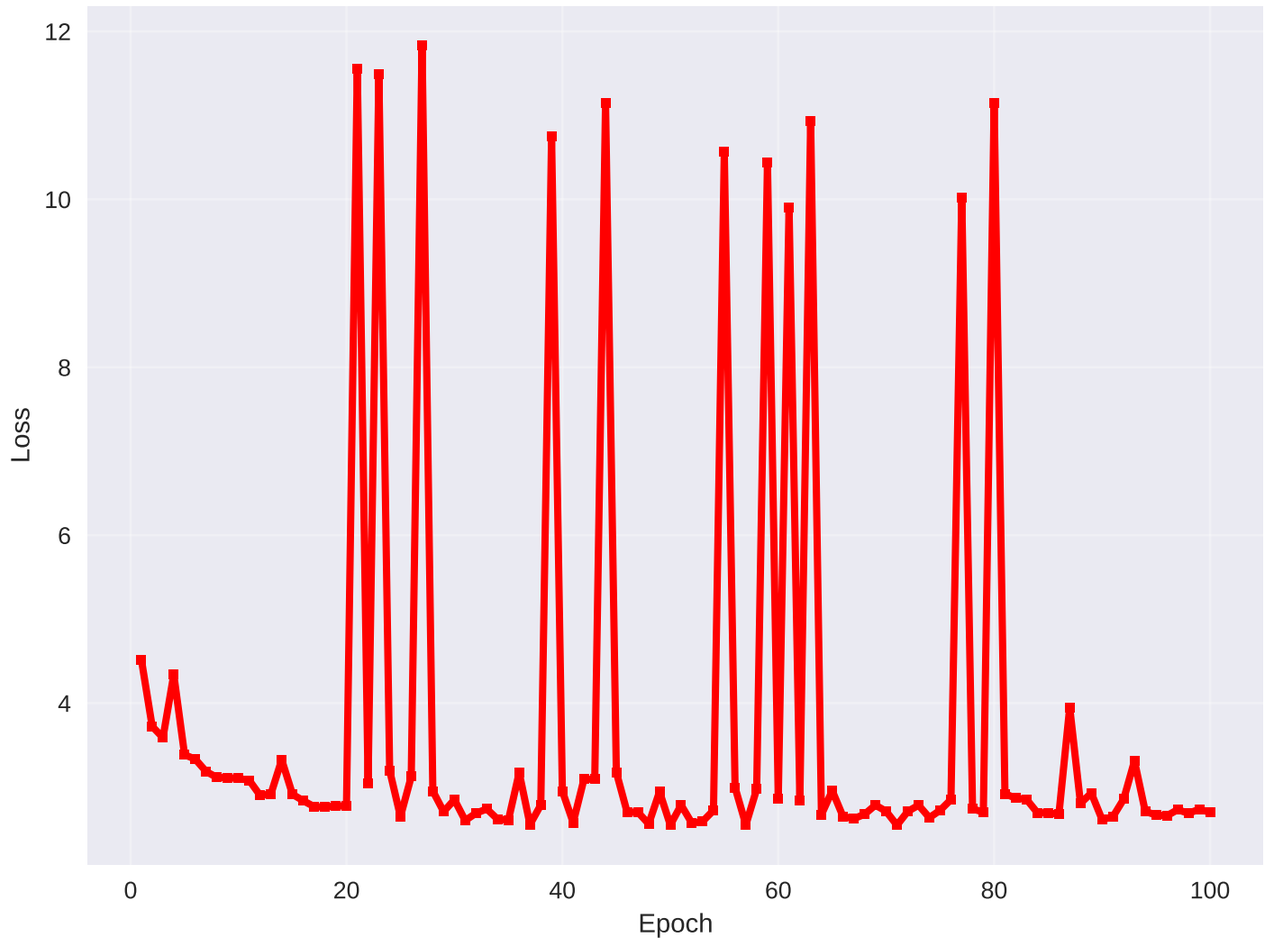
Federated Learning - Main Metrics Overview

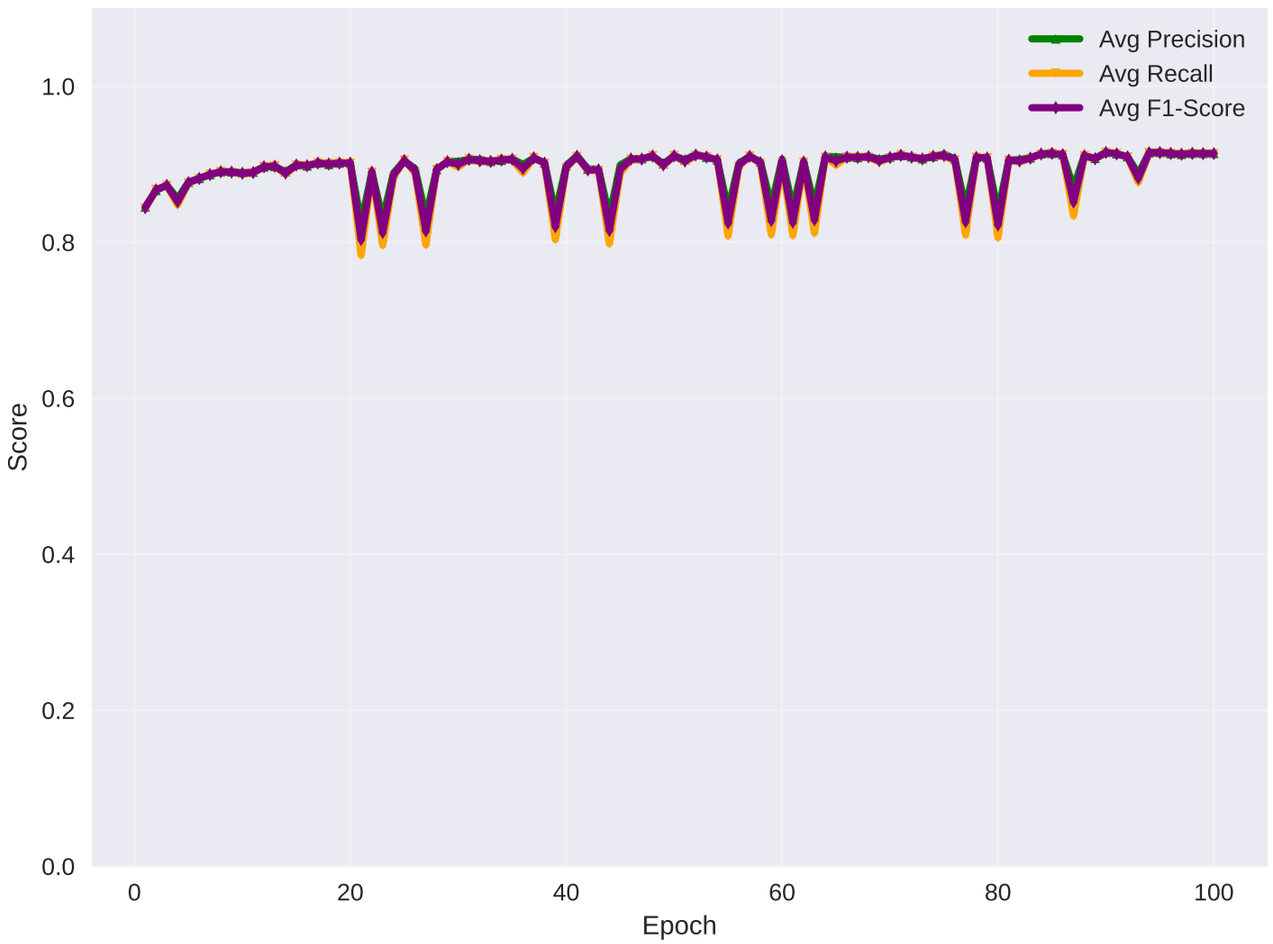
 Global Model Accuracy Evolution



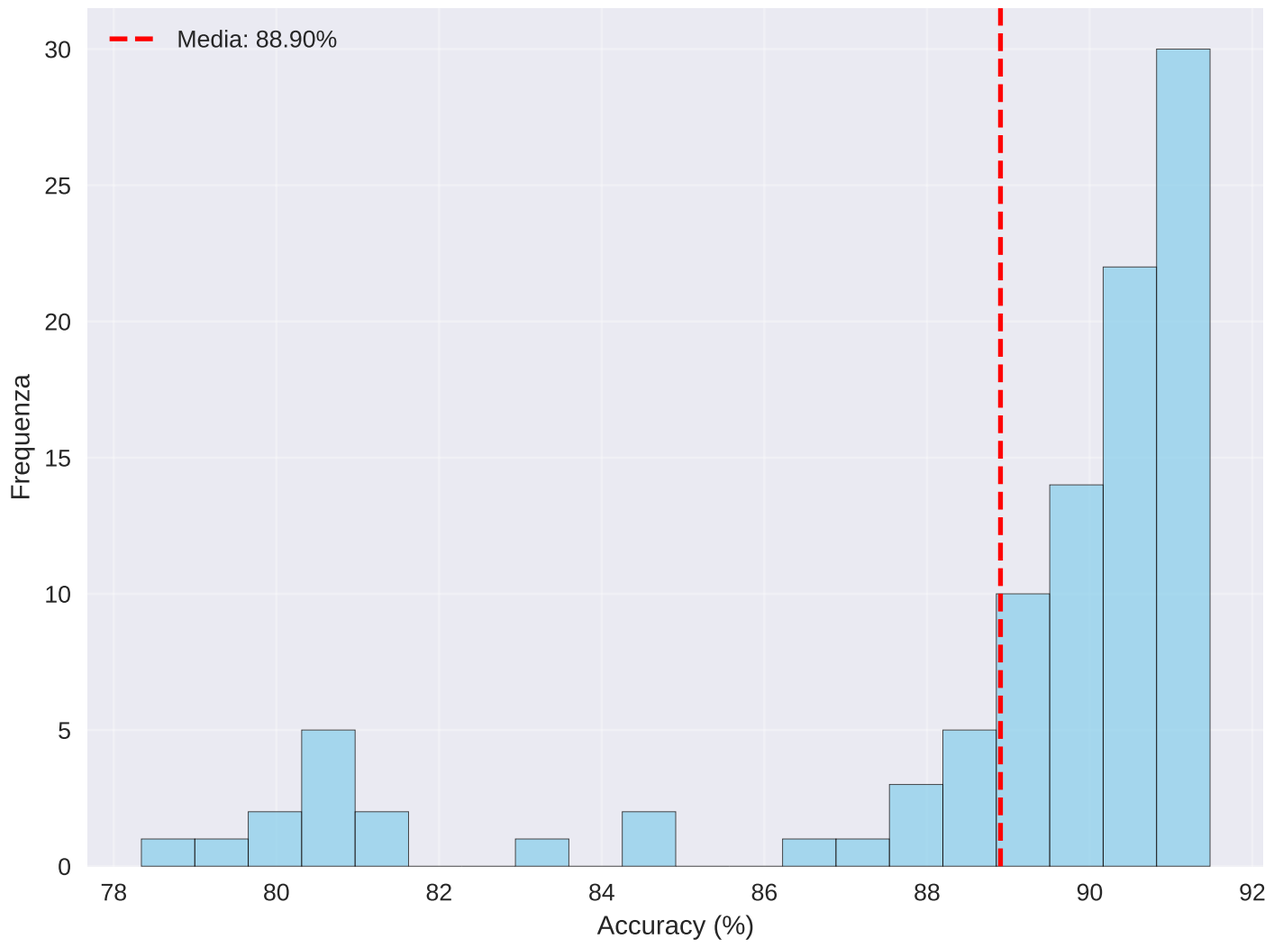
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score



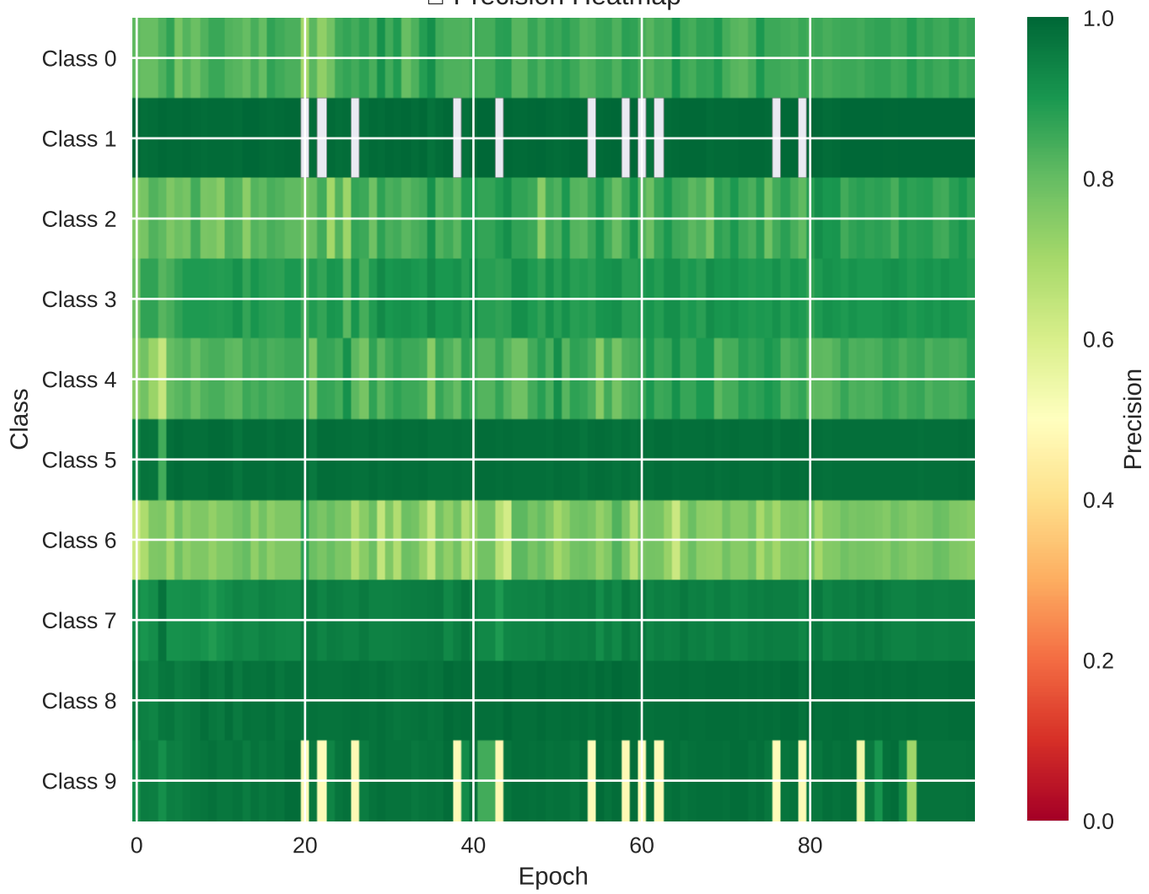
 Distribuzione Accuracy

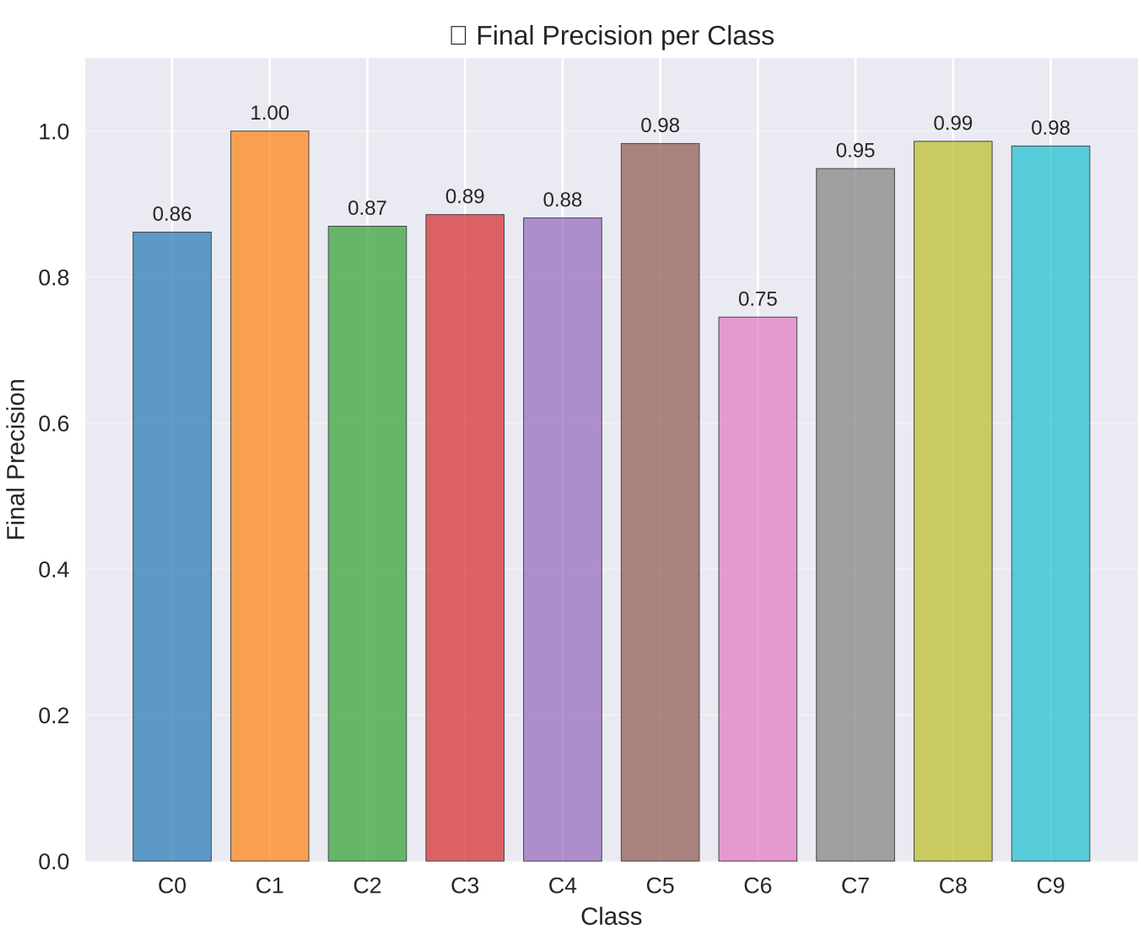


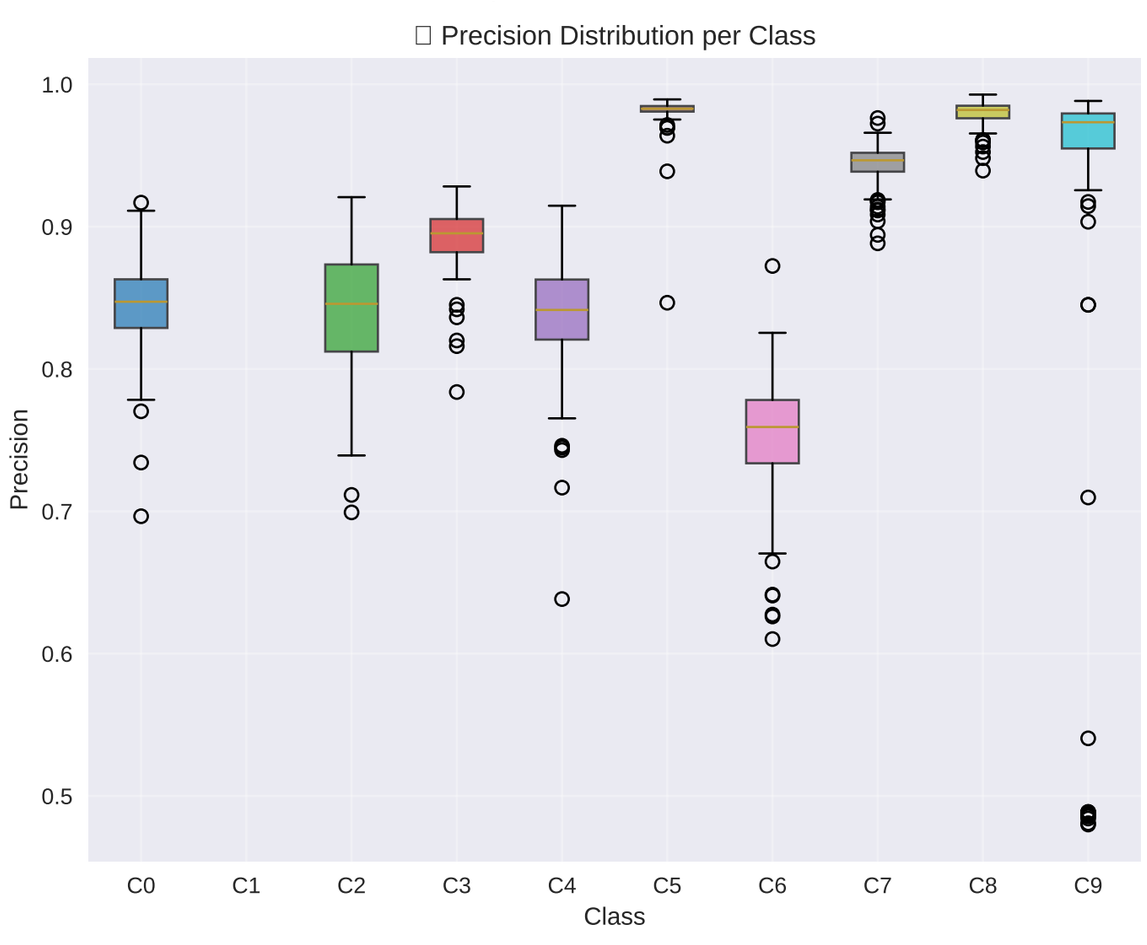
Per-Class Precision Analysis





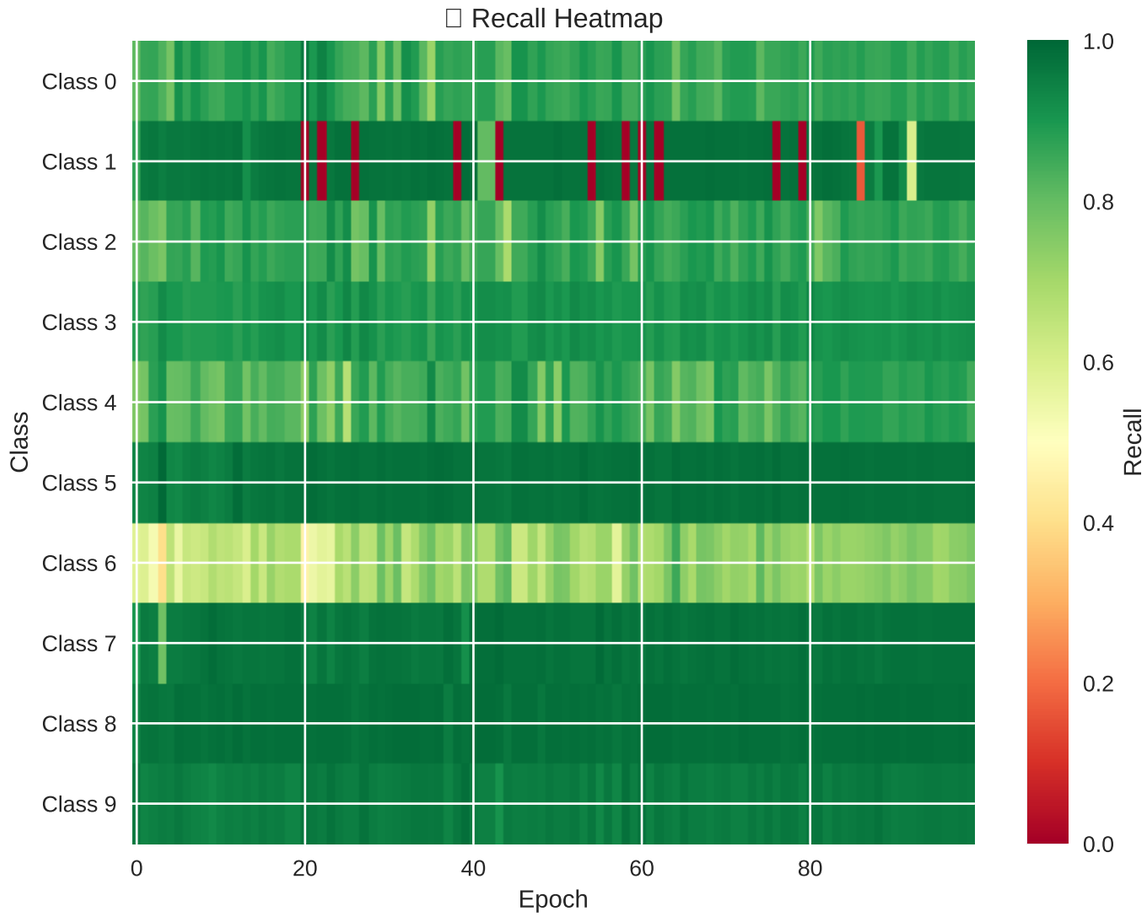


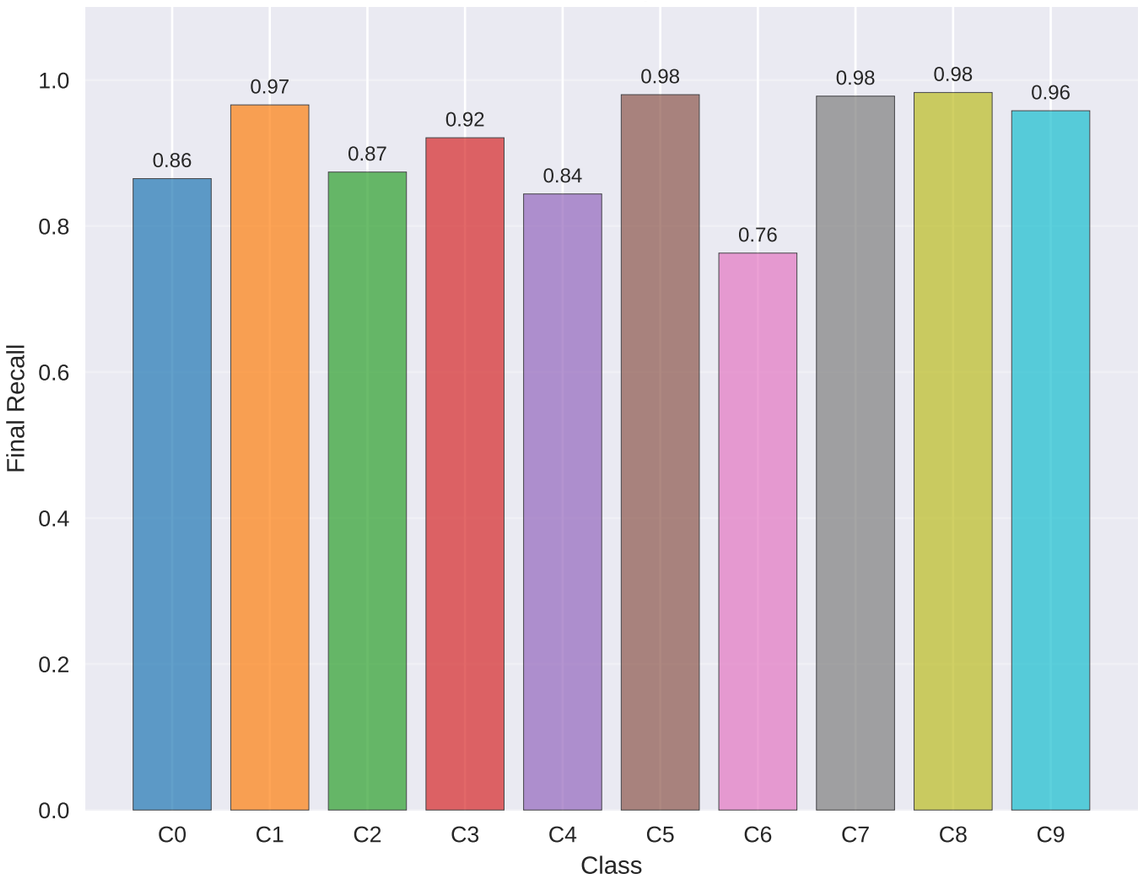


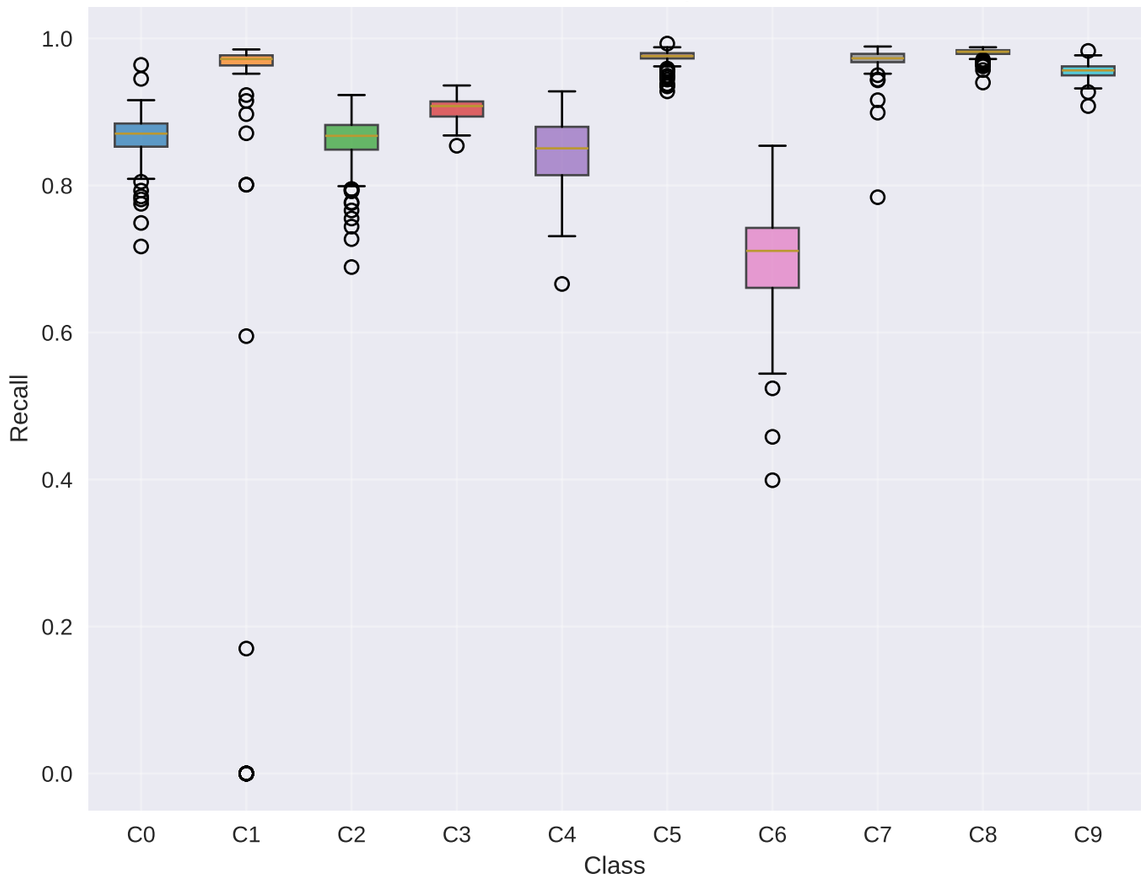


Per-Class Recall Analysis

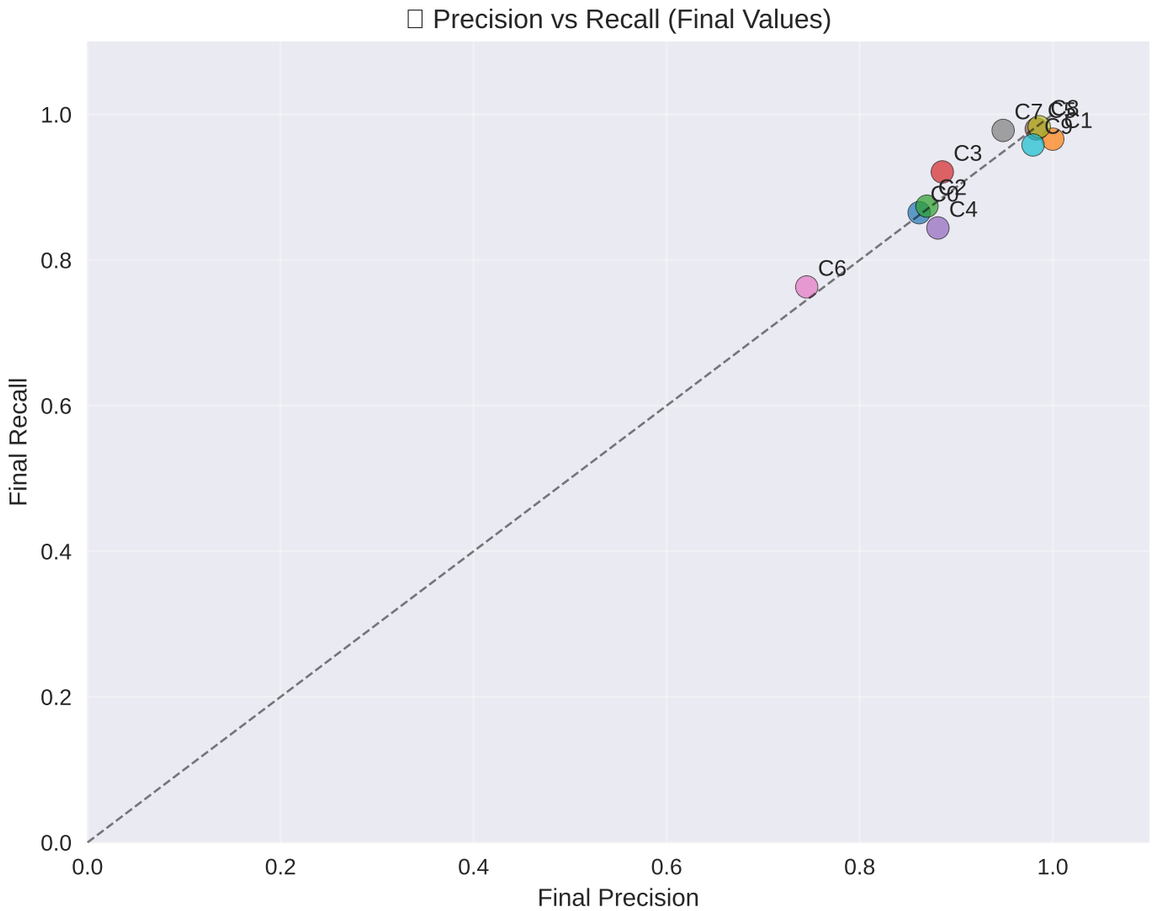




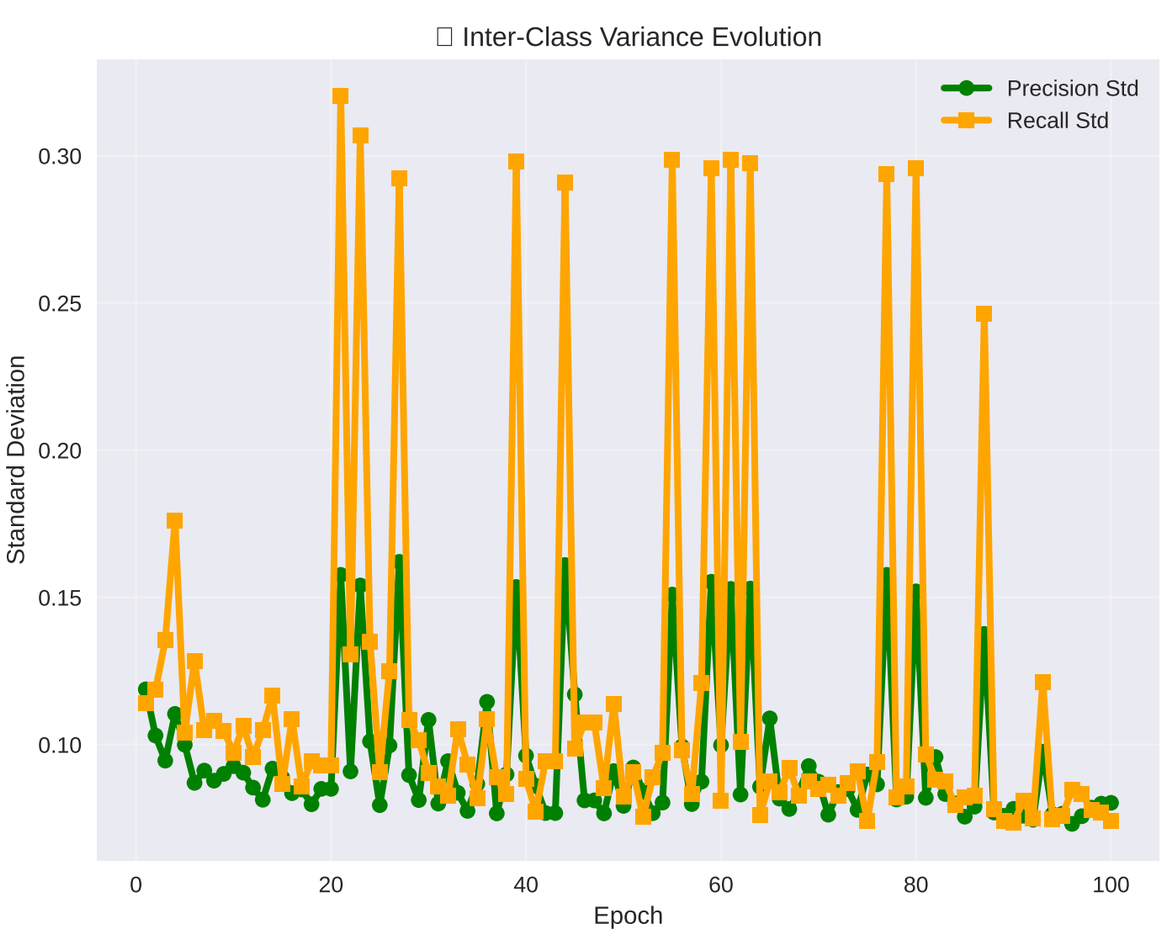


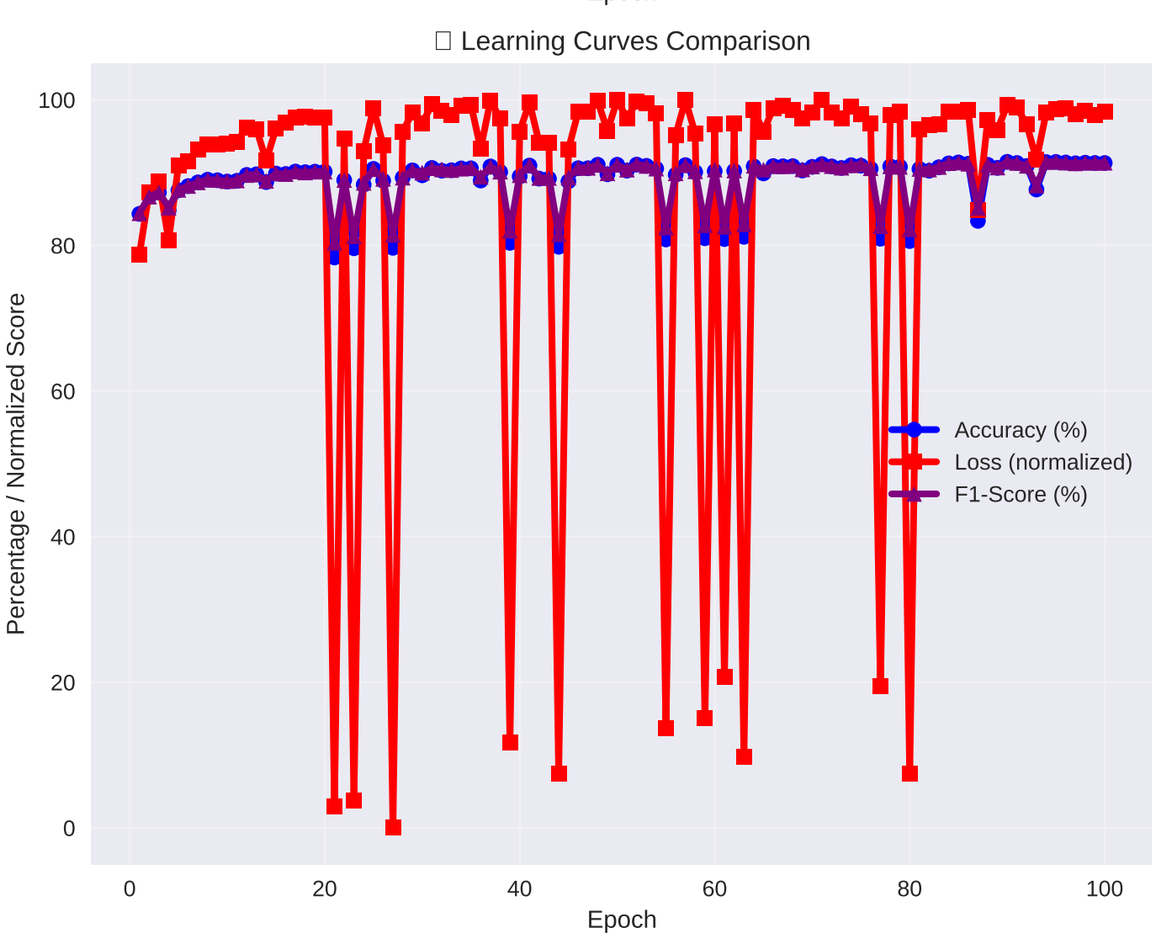


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-01 12:29:55
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.32%
• Loss: 2.6991
• Average Precision: 0.914
• Average Recall: 0.913
• Average F1-Score: 0.914

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.99%
• Riduzione Loss: +1.8194
• Stabilità Accuracy (CV): 0.0379

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (1.000)
• Peggiore Precision: Classe 6 (0.745)
• Migliore Recall: Classe 8 (0.983)
• Peggiore Recall: Classe 6 (0.763)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.862, R=0.865, F1=0.863
• Classe 1: P=1.000, R=0.966, F1=0.983
• Classe 2: P=0.870, R=0.874, F1=0.872
• Classe 3: P=0.886, R=0.921, F1=0.903
• Classe 4: P=0.881, R=0.844, F1=0.862
• Classe 5: P=0.983, R=0.980, F1=0.981
• Classe 6: P=0.745, R=0.763, F1=0.754
• Classe 7: P=0.949, R=0.978, F1=0.963
• Classe 8: P=0.986, R=0.983, F1=0.984
• Classe 9: P=0.980, R=0.958, F1=0.969

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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